newleaf

New page. Nothing lost.

for Claude Code

−86%

tokens per task

measured, same task, same quality

THE GUIDE

Fresh Claude Code sessions that keep the thread

newleaf writes a handoff note in one call, loads it into your next session on its own, and tells you when a chat has grown costly. Starting fresh stops costing you your context, so you can do it often, and stop paying to re-send a giant conversation on every step.

−86%
tokens per task, measured on the same task with the same quality

1 call
to write the handoff note, saved by a hook

0 setup
works in every project, from any terminal

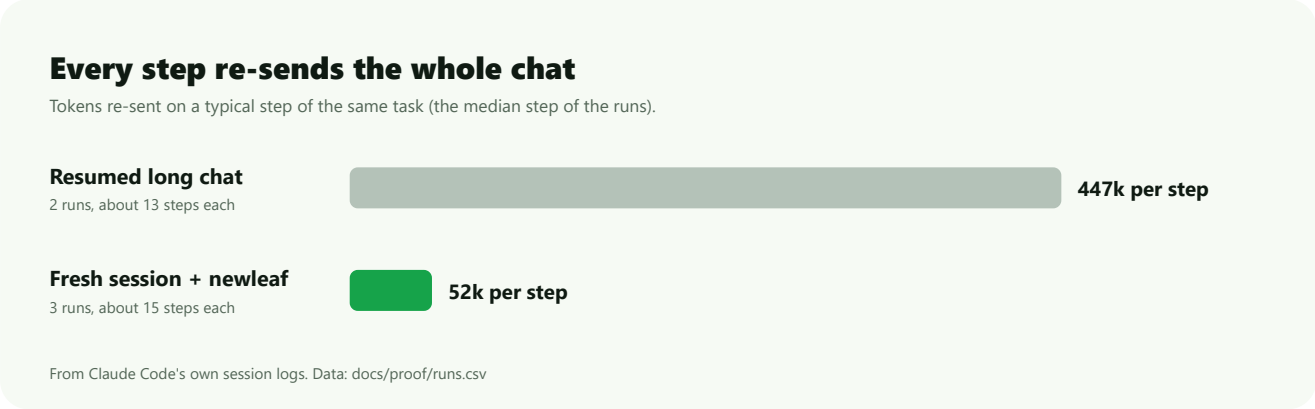
Inside: why your limits burn (page 2), how newleaf works (3), installing and using it (4), the proof and how it was measured (5), and answers to common questions (6).

Version 1.0.0 · 2026-10-31 · MIT licence
github.com/zannn123/newleaf

THE PROBLEM

Why your limits burn

A Claude Code session is a conversation, and every step of it (each tool call, each answer) sends the whole conversation to the model again. A cache makes the re-sending cheaper, but it still counts. Keep one session open for days and each step drags hundreds of thousands of tokens along, so even a one-line change costs millions.



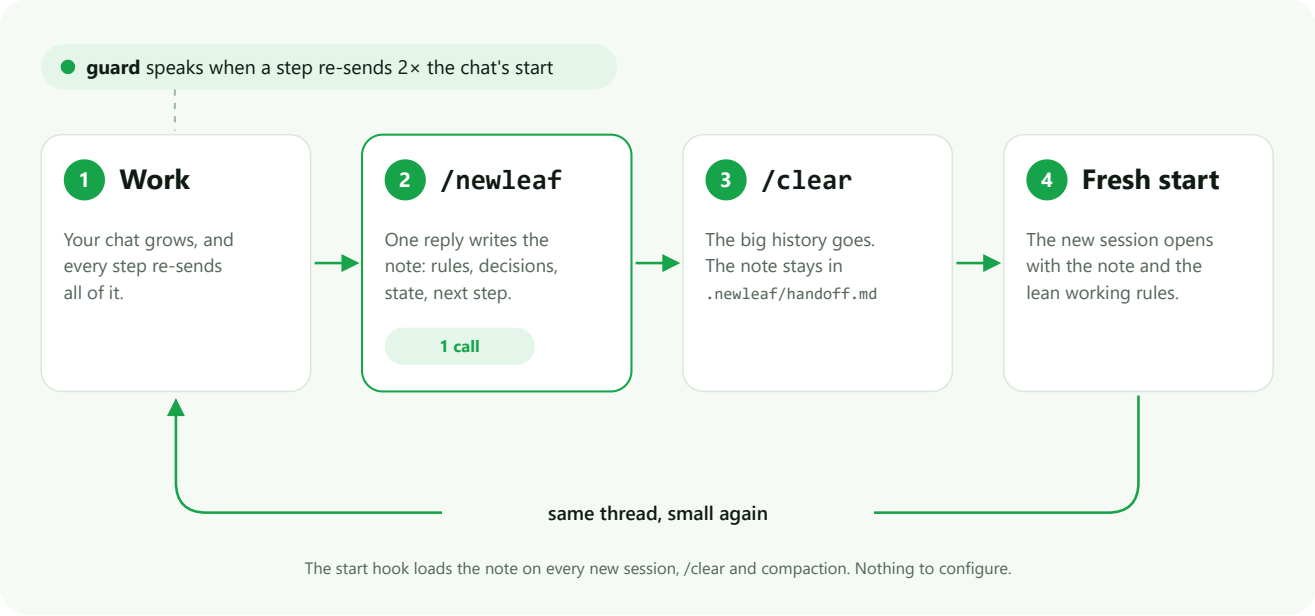
What the logs said

- 14 days of real Claude Code logs, 8,030 API calls: **99%** of all tokens were the conversation being re-sent; the model's replies were **0.3%**.
- The long-running session re-sent a median of **548k** tokens per call; a fresh session started near **58k**.
- A small request (five calls or fewer) in that session cost a median of **1.5 million** tokens.

So writing shorter replies cannot save much. What saves is a smaller conversation. People keep long sessions because a fresh one forgets the rules and decisions they set up. newleaf removes that reason.

HOW IT WORKS

A loop of four steps



The start hook loads the note on every new session, /clear and compaction. Nothing to configure.

PIECE	WHAT IT DOES
`/newleaf`	One reply writes the note. The branch, commit, changed files and recent commits are gathered before the model reads the skill, so the note costs a single call.
Stop hook	Takes the note from the reply and saves it as this session's own in `.newleaf/notes/` and as the folder's latest, `.newleaf/handoff.md`, and adds a `.gitignore` so the folder never reaches git.
Start hook	Loads the lean working rules into every session. `/clear` and compaction also get that session's own note, and hear if the branch or commit moved on. A brand-new session gets one line pointing at the latest note.
Guard	Before each prompt, compares what a step re-sends now with the chat's first step. At 2× it says so, again at 4× and 8×. A handoff costs one call, so past 2× it pays back within the next request.

A note looks like this

```
---
written: 2026-10-31T18:42:00+0800
branch: feature/checkout
head: 9f3c2a1
---
## Rules in play
- "No hard-coded colours: use the theme tokens."
## Decided
- The split button keeps its arrow menu; only its style changes.
## State
- Done: line style on both halves, report tests pass. Not committed.
## Next
- Commit, then update the mockup.
## Open
- Should "Excel" stay first in the menu?
```

INSTALL AND USE

Two commands, then three keystrokes a round

```
# in any terminal: PowerShell, cmd, bash, zsh
claude plugin marketplace add zannn123/newleaf
claude plugin install newleaf@newleaf
```

Or inside Claude Code: `/plugin marketplace add zannn123/newleaf`, then `/plugin install newleaf@newleaf`. It applies to every project from then on.

Needs: bash. It is built in on macOS, Linux and WSL; on Windows it comes with Git for Windows, which Claude Code recommends. No Python, no Node, no network.

Every round of work

1. Work as usual.
2. When the round ends, or the guard speaks, type `/newleaf`. One reply writes the note and the hook saves it.
3. Type `/clear`. The fresh session starts with the note and the rules.

The guard's line looks like this:

```
newleaf: every step of this chat now re-sends 612k tokens, 10x its start (59k).
Type /newleaf, then /clear: the next session starts small with the note.
```

Settings (optional)

`~/.claude/newleaf.json` for your machine, or `.newleaf/config.json` in a project; the project wins.

KEY	DEFAULT	MEANING
`rules`	`true`	`false` leaves out the lean working rules.
`warn_at`	2× the chat's first step	Tokens per step that wake the guard; `0` turns it off.

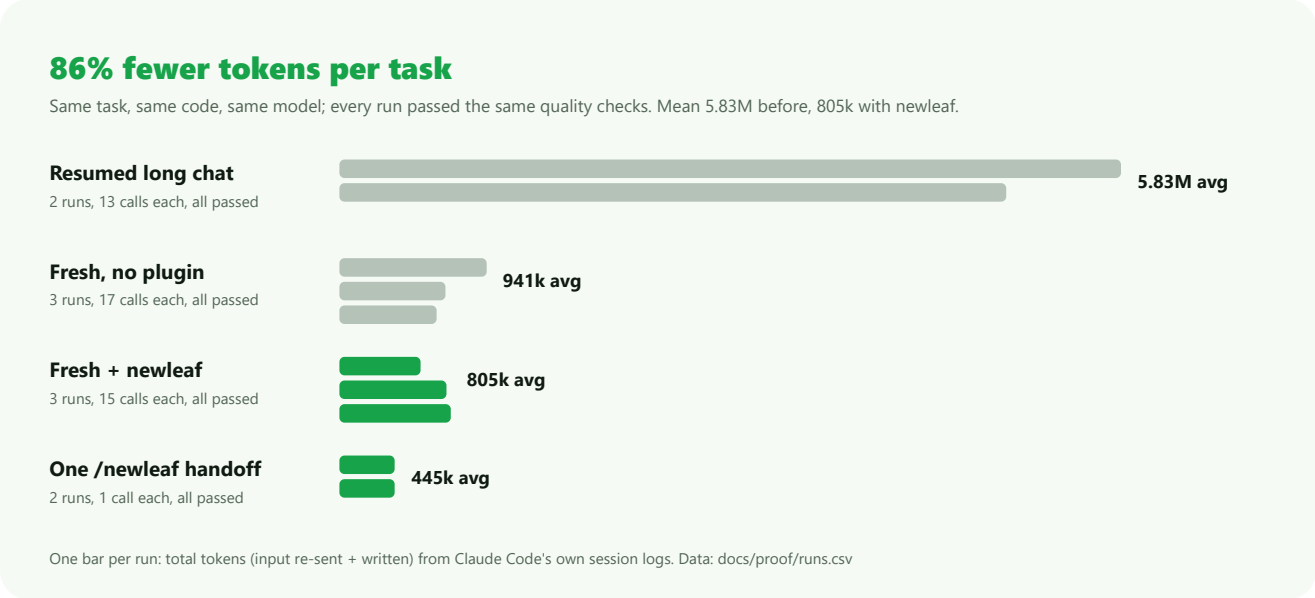
Remove

```
claude plugin uninstall newleaf
claude plugin marketplace remove newleaf
```

PROOF

Measured, not estimated

One real task on a real production Flutter codebase: restyle a split button. Same starting code, same prompt, same model (Claude Opus). Tokens come from Claude Code's own session logs: everything each call re-sent plus everything it wrote. Every run had to pass the same checks: the project's 97 report tests, the analyzer, no hard-coded colour, and changes only inside the feature's folder.



CONDITION	RUNS	TOKENS PER TASK	CALLS	RE-SENT PER CALL	QUALITY
Resumed long chat (before)	2	5.83M	13	447k	2 / 2
Fresh session, no plugin	3	941k	17	57k	3 / 3
Fresh session + newleaf	**3**	**805k**	**15**	**52k**	**3 / 3**
One `/newleaf` handoff	2	445k	1	441k	2 / 2 notes

- **86% fewer tokens per task.** 79% fewer on the first task even counting the handoff; 85% over five tasks.
- "Before" resumed a real session at the moment its owner paused work, 440k tokens of history, exactly as one would resume it.
- Limits: one codebase, one task type, headless runs. Your numbers will differ; `tools/measure.py` shows yours.

QUESTIONS

Answers

Does it work with caveman?

Yes, they add up. caveman shortens replies, about 0.3% of tokens; newleaf cuts the re-sent conversation, about 99%.

Where is my note?

The latest is `.newleaf/handoff.md` in the project, each session's own in `.newleaf/notes/`, so sessions side by side keep their notes apart. Git ignores the folder on its own.

Will a fresh session forget my rules?

Not the ones in the note: "Rules in play" carries them in your words, and the start hook loads the note before your first prompt.

Is anything sent anywhere?

No. The hooks read your local session log and write `.newleaf/` in your project. No telemetry.

The guard speaks too soon

Set `warn_at` to the tokens per step you accept, or `0` to switch it off.

Nothing happens on Windows

The hooks need bash: install Git for Windows. Then `claude plugin list` should show `newleaf` enabled.

How do I check it works?

Clone the repo and run `bash tests/hooks.sh` (every hook, offline) and `bash tests/smoke.sh` (end to end with your Claude Code).

Where was it tested?

Windows 11 (Git Bash), Ubuntu, Debian (mawk), Alpine (busybox awk) and macOS (CI).

Does it work with Codex?

Experimental: `codex plugin marketplace add zannn123/newleaf`, `codex plugin add newleaf@newleaf`, trust its hooks with `/hooks`, then `$newleaf`, `/new`, and "continue from the note". Built from Codex's docs and source; not yet measured in a live Codex session.